**11** In May 2019, the distance between Earth and Mars was $3.9 \times 10^7$ km.

In May 2019, a signal was sent from Earth to Mars.
Assuming that the signal sent from Earth to Mars travelled at a speed of $3 \times 10^5$ km per second,

(a) how long did the signal take to get to Mars?

Time = distance ÷ speed

= $(3.9 \times 10^7) \div (3 \times 10^5)$ = 130 s

**1 mark**
Use distance ÷ speed.

**1 mark**
130 seconds.

130 seconds

**(2)**

The speed of the signal sent from Earth to Mars in May 2019 was actually less than $3 \times 10^5$ km per second.

(b) How will this affect your answer to part (a)?

At lower speed the journey takes longer.

**1 mark**
Actual time greater than 130 s.

**(1)**

**(Total for Question 11 is 3 marks)**